

BRIDGE L2194
Carries 200th Avenue over unnamed stream
Magnolia
Rock County
Minnesota

HAER MN-130
HAER MN-130

PHOTOGRAPHS

WRITTEN HISTORICAL AND DESCRIPTIVE DATA

HISTORIC AMERICAN ENGINEERING RECORD
National Park Service
U.S. Department of the Interior
1849 C Street NW
Washington, DC 20240-0001

HISTORIC AMERICAN ENGINEERING RECORD

BRIDGE L2194

HAER No. MN-130

Location: Bridge L2194 is located in Magnolia Township, Rock County, Minnesota, and carries a gravel road, 200th Avenue, over an unnamed stream.

Bridge L2194 is located at latitude 43.631286, longitude -96.073569. The coordinates represent the center of the structure. They were obtained in 2016 by plotting the bridge's location in Google Earth. The location has no restrictions on its release to the public.

Present Owner: Magnolia Township.

Present Use: This bridge currently carries vehicular traffic over an unnamed stream.

Significance: Bridge L2194 is locally significant under National Register of Historic Places (National Register) Criterion C in the area of Engineering as an example of the work of local bridge builder, Perley N. Gillham. The bridge is noted as being significant for displaying characteristic elements of Gillham's design style, including decorative coping and molding and solid railings with cylindrical rail posts. The bridge is also significant for being a distinctive example of Gillham's box culvert designs. This bridge is the last known Gillham bridge to be designed and constructed, and illustrates the full extent of his career in Rock County.

Historians: Kelli Andre Kellerhals and Saleh Miller, The 106 Group Ltd.; 2017.

Project

Information: This study, which is the third phase in the Minnesota Department of Transportation's (MnDOT) multi-phased Local Historic Bridge Study, was developed and executed in partnership with representatives from the Minnesota Historic Preservation Office (MnHPO); Federal Highway Administration (FHWA); MnDOT's State Aid for Local Transportation (SALT); the Corps of Engineers (Corps); local public works and highway departments; county and township boards, and city councils; the preservation community; and the general public. MnDOT's SALT is the project lead and is jointly administering the program with MnDOT's Cultural Resources Unit (CRU) and Bridge Office. This study prepared Historic American Engineering Record (HAER) documentation for bridges that are unlikely to be rehabilitated; are rare, unusual, and the first or only

known example; or that are candidates for relocation. Documentation is being prioritized for bridges in a state of severe deterioration or subject to loss. The Minnesota Department of Transportation retained LHB, Inc.; Mead & Hunt, Inc.; and The 106 Group Ltd. to prepare this documentation. Kelli Andre Kellerhals and Saleh Miller wrote the history. Dietrich Floeter completed the photography.

PART I. HISTORICAL INFORMATION

A. PHYSICAL HISTORY

Date(s) of Construction: 1928.

Architect/Engineer: Perley N. Gillham.

Builder/Contractor/Supplier: Unknown.

Original plans and construction: Original plans, perspectives, or historical photographs for this bridge are not available.

Alterations and additions: Records do not exist that document any changes to this bridge over time. However, a visual inspection undertaken in 2014 indicates that the cell walls appear to have been extended on either side of the bridge, as the original concrete shows impressions from wood plank forms while the extensions show markings of plywood forms.

B. HISTORICAL CONTEXT

According to MnDOT records, Bridge L2194 was built in 1928.¹ Research at the City of Luverne, the Rock County Courthouse, the Rock County Highway Department, and the Rock County Historical Society did not yield any information on this bridge's construction. However, the "Gillham" inscription on the south railing confirms that this bridge was constructed by local bridge builder Perley N. Gillham. Additionally, records of any changes, alterations, or repairs to this bridge have not been found. The lack of information regarding the construction and maintenance of this bridge is to be expected, as the "Reinforced-Concrete Highway Bridges in Minnesota, 1900-1945" Minnesota Multiple Property Documentation Form (MPDF) notes that "there is very little documentation of reinforced-concrete bridge construction in Minnesota for the years prior to state involvement."² Therefore, "almost all the evidence exists in the few surviving structures themselves."³

¹ Minnesota Department of Transportation [MnDOT], "MnDOT Structure Inventory Report: Bridge L2194" (Saint Paul: Minnesota Department of Transportation, 2012).

² Robert Frame, "Reinforced-Concrete Highway Bridges in Minnesota, 1900-1945" (Saint Paul: Minnesota State Historic Preservation Office, 1988), E-11.

³ Frame, "Reinforced-Concrete Highway Bridges," E-11.

Bridge L2194 was one of a number of bridges designed and built by Perley Nye Gillham. Despite several studies on Gillham bridges over the years, very little is actually known about him, other than that he was “an obscure mason and general contractor” who designed and built “small but elegant reinforced-concrete bridges in Rock County.”⁴ Gillham was born in Jamestown, Wisconsin, on November 25, 1854.⁵ He arrived in Luverne, Minnesota, in 1875, following his brother Edwin, who had arrived in 1868 and worked as a mail carrier on a stage coach route.⁶⁷ The National Register nomination for Bridge L2162, an extant reinforced-concrete arch bridge by Gillham also located in Rock County, summarized what little is known about him:

He first appears in county commissioners’ records for plastering work in 1875, and periodically is mentioned in connection with various contracting work. His name is first associated with a bridge, but only for repairs, in 1883. In 1887 he was appointed superintendent of construction for the new county courthouse, and in 1900 was awarded the contract for construction of a new county jail. From the 1890s onward, he [sic] name appears regularly for miscellaneous county work, including building, bridge, and road contracts. Unfortunately, the county commissioner’ minutes rarely discuss bridge work in any detail, and usually do not mention the bridge type or the contractor’s name. A 1934 county history referred to Gillham as ‘our first contractor and builder,’ noting that ‘many of the prominent buildings of the city at the present time were built by him.’⁸

Further details on Gillham’s life come from an December 27, 1945, obituary that states, “P. N. Gillham, 91, long-time resident of Luverne, died on December 14, at the home of his son near Tucson, Ariz.”⁹ The obituary also notes that “Mr. Gillham will be remembered as bridge contractor in Rock county [sic]. He also made many bridges in adjoining Pipestone and Nobles counties.”¹⁰ The obituary further recounts Gillham’s 45-year career, which began in 1883, at the age of 29, when he was awarded a contract to repair the Ash Creek Bridge in Rock County, and continued until he constructed his last bridge in 1928, at the age of 74.¹¹

The bridges Perley Gillham designed and built in Rock County are easily identifiable by their shared common characteristics. Gillham bridges are almost exclusively single span, reinforced-concrete, arch

⁴ Frame, “Reinforced-Concrete Highway Bridges,” E-11.

⁵ Arizona State Department of Health, “Standard Certificate of Death: Perley Nye Gillham,” (Phoenix: Arizona Department of Health Services), State File 529, Registrar’s No. 1143.

⁶ Robert Frame, “National Register of Historic Places Nomination (NRHP) Form: Bridge L-2162,” (Saint Paul: Minnesota State Historic Preservation Office, 1988), 8-2.

⁷ “P. N. Gillham Taken By Death,” *Rock County Star*, December 27, 1945.

⁸ Frame, “NRHP Form: Bridge L-2162,” 8-2.

⁹ “P. N. Gillham Taken By Death,” *Rock County Star*.

¹⁰ “P. N. Gillham Taken By Death,” *Rock County Star*.

¹¹ “P. N. Gillham Taken By Death,” *Rock County Star*.

bridges with closed spandrels, found in remote locations throughout the county. These small, simple structures display some decorative elements, suggestive of the classicism of the City Beautiful Movement that flourished around the turn of the century. Moldings are often used to delineate the deck and the railings, while the railings, which are usually solid, often have cylindrical end posts.¹² Bridge L2194 is somewhat unique amongst Gillham's body of work as it is a concrete box culvert design, which is a design not frequently used by Gillham.¹³

Perley Gillham is noteworthy for his early use of reinforced-concrete for bridge construction. It is not known exactly how Gillham came to build reinforced-concrete bridges during the earlier years of this construction type. In the "Reinforced-Concrete Highway Bridges in Minnesota, 1900-1945" MPDF, author Robert Frame suggests that the 1894 Fritz von Emperger-designed Melan-arch concrete bridge in Rock Rapids, Iowa, may have been influential in Gillham's work (see Melan Arch Bridge, HAER IA-15). Rock Rapids is located only a few miles from Luverne and it is possible that he traveled to see this bridge.¹⁴ The proportions and dimensions of the Rock Rapids Bridge are almost identical to Gillham's designs for reinforced-concrete arch bridges. It has also been suggested that Gillham may have had an opportunity to develop a relationship with noted Minneapolis bridge builder William S. Hewett. In 1883, Gillham repaired the Ash Creek Bridge in Rock County and in 1884 Hewett was awarded the contract for the replacement of this same bridge.¹⁵ While all such suggestions of influence are speculative, these potential connections serve to show that Gillham did not necessarily work in isolation and was likely well aware of current design methods and ideas.

While Gillham's obituary indicates that he designed and built bridges in several counties in southwestern Minnesota, only one bridge in Nobles County and no bridges in Pipestone County have been identified.¹⁶ The greatest concentration of his work is in Rock County, where he is estimated to have constructed about 90 percent of the concrete bridges in the county.¹⁷ Although the exact number of bridges that Gillham designed and built is unknown, at least 12 bridges in Rock County and one in Nobles County are known to have been attributed to him through the identification "P. N. Gillham" inscribed into the concrete. As noted in the National Register nomination for Bridge L2162, "There are at least 16 additional bridges in the county that do not have an inscription, but have design elements so

¹² 106 Group, "27 Bridges in Rock County," 13.

¹³ Minnesota Department of Transportation [MnDOT], "Minnesota Historic Highway Bridge Inventory: Bridge No. L2194" (Saint Paul: Minnesota Department of Transportation, 1994).

¹⁴ Frame, "Reinforced-Concrete Highway Bridges," E-11.

¹⁵ Frame, "NRHP Form: Bridge L-2162," 8-2.

¹⁶ "P. N. Gillham Taken By Death," *Rock County Star*; The 106 Group, Ltd. "Evaluation of 27 Bridges in Rock County, Minnesota for the National Register of Historic Places," (Saint Paul: The 106 Group, 1994), 14.

¹⁷ "P. N. Gillham Taken By Death," *Rock County Star*.

similar to the confirmed Gillham bridges as to allow attribution to him.”¹⁸ Bridges attributed to Gilliam based on an inscription span the years 1908-1913, while bridges attributed to him based on design characteristics span the period c. 1901-1926.¹⁹

The earliest known bridge designed by Gillham was Bridge L2315 (RK-KAN-004) in Kanaranzi Township, designed ca. 1901; according to SHPO records this bridge has been razed. Constructed in 1928, Bridge L2194 (RK-MGT-002) holds the distinction of being the last known bridge to have been designed and built by Gillham during his long career. According to a 2014 review of extant Gillham bridges, only four known and attributed bridges in Minnesota remain. These include:

- Bridge L2257 (1910; RK-LVC-032, see HAER MN-131)
- Bridge 2340 (1906; RK-BCT-005)
- Bridge L4646 (1911; RK-BCC-003)
- Bridge L2194 (1928; RK-MGT-002)²⁰

Bridge L2182 was also designed by Gillham but is nonextant (see HAER MN-79).

PART II. STRUCTURAL/DESIGN INFORMATION

A. GENERAL STATEMENT

Character: Bridge L2194 is exemplary of the work of local bridge builder, Perley N. Gillham. The bridge is significant for displaying characteristic elements of Gillham’s design style, including decorative coping and molding and solid railings with cylindrical end posts. The bridge is also significant for being a distinctive example of Gillham’s box culvert designs. The bridge is the last known designed and constructed by Gillham, and illustrates the full extent of his career in Rock County.²¹

Condition of fabric: Much of this bridge is in fair condition, with some elements in poor condition. Bridge Inspection Reports note some cracks in the concrete railing and headwalls which has allowed

¹⁸ Frame, “NRHP Form: Bridge L-2162,” 8-1.

¹⁹ 106 Group, “27 Bridges in Rock County,” 14.

²⁰ LHB, Corp. and Mead & Hunt, Inc. “Local Historic Bridge Report – Abridged” for Bridge L2194 prepared for the Minnesota Department of Transportation (July 2014).

²¹ MnDOT, “Minnesota Historic Highway Bridge Inventory: Bridge No. L2194;” Bridge L2194 is significant within the “Reinforced-Concrete Highway Bridges in Minnesota Multiple Property Documentation Form (MPDF)” (Frame, 1988), under Registration Requirement 6 for association with an important Minnesota builder.

moisture to penetrate rebar causing rusting with a loss of section. There are also areas of the bridge that are spalling, which has exposed the rebar. Scour at the inlet has begun to undermine the box as well.²²

B. DESCRIPTION

This bridge is a small, two-span, reinforced-concrete box culvert with a total structure length of 18 feet and a deck width of 18'-7".²³ The substructure is comprised of two cast-in-place concrete cells that are each approximately 8' wide and have a barrel length of 21'-0". In an apparent alteration, the cell walls extend out beyond the width of the culvert and form the wing walls on the north and south ends of the bridge. The cell walls also create a center pier that extends out beyond the width of the culvert on both sides. This also appears to be an alteration. The wing walls slightly angle out and downward from the approach. The deck has a gravel surface. On the outside face of the bridge, the deck is distinguished by a band of decorative coping that is surmounted by two-foot tall, solid concrete railings that feature cylindrical end posts which extend below the deck and are integrated into the wing walls. The inside faces of the end posts are flush with and integral to the railing. The railing and end posts are capped with decorative coping and molding. The south railing has a partial visible inscription which reads "Gillham."

C. SITE INFORMATION

This bridge is located in Magnolia Township, Rock County, Minnesota. The bridge has a north to south alignment and carries a gravel road, 200th Avenue, over an unnamed stream. The bridge is located in a rural, agricultural area surrounded by farm fields.

PART III. SOURCES OF INFORMATION

A. PRIMARY SOURCES

Arizona State Department of Health, "Standard Certificate of Death: Perley Nye Gillham." Phoenix: Arizona Department of Health Services. State File 529, Registrar's No. 1143.

Minnesota Department of Transportation. "Minnesota Historic Highway Bridge Inventory: Bridge No. L2194." St. Paul: Minnesota Department of Transportation, 1994.

Minnesota Department of Transportation. "MnDOT Structure Inventory Report: Bridge No. L2194." St. Paul: Minnesota Department of Transportation, 2012.

"P. N. Gillham Taken By Death." *Rock County Star*, December 27, 1945.

²² MnDOT, "Structure Inventory Report: Bridge L2194."

²³ MnDOT, "Structure Inventory Report: Bridge L2194."

B. SECONDARY SOURCES

Frame, Robert. "National Register of Historic Places Multiple Property Documentation Form: Reinforced-Concrete Highway Bridges in Minnesota." Saint Paul: Minnesota Historic Preservation Office, 1988.

Frame, Robert. "National Register of Historic Places Nomination Form: Bridge L-2162." Saint Paul: Minnesota Historic Preservation Office, 1988.

LHB, Corp. and Mead & Hunt, Inc. "Local Historic Bridge Report – Abridged" for Bridge L2194. Prepared for the Minnesota Department of Transportation (July 2014).

The 106 Group Ltd. "Evaluation of 27 Bridges in Rock County, Minnesota for the National Register of Historic Places." Saint Paul: The 106 Group Ltd, 1994.